

ADVANCED LINEAR ALGEBRA

HOMEWORK 1

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P.A.1 Show that:

- (a) $(1 + i)^4 = -4$
- (b) $(1 - i)^{-1} - (1 + i)^{-i} = i$
- (c) $(i - 1)^{-4} = -1/4$
- (d) $10(1 + 3i)^{-1} = 1 - 3i$
- (e) $(\sqrt{3} + i)^3 = 8i$

Solution. Solution P.A.1

□

P.A.3 Let $z = 3 + 4i$. Plot and label the following points in the complex plane: z , $-z$, $\bar{z}$, $-\bar{z}$, $1/z$, $1/\bar{z}$, $-1/\bar{z}$

Solution. Solution P.A.3

□

P.A.6 Write the following in polar form:

- (a) $1 + i$
- (b) $(1 + i)^2$
- (c) $(1 + i)^4$

Solution. Solution P.A.6

□

P.A.7 What are the square roots of $1 + i$? Draw a picture of their locations.

Solution. Solution P.A.7

□

P.A.11 If $z \neq 0$, show that $\overline{1/z} = 1/\bar{z}$.

Solution. Solution P.A.11

□

P.B.2 If p is a real polynomial, show that $p(\lambda) = 0$ if and only if $p(\bar{\lambda}) = 0$.

Solution. Solution P.B.2

□

P.C.3 If $A \in \mathbf{M}_n(\mathbb{F})$ is invertible, show that $A^\top$ is invertible and $(A^{-1})^\top$ is its inverse.

Solution. Solution P.C.3

□

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P.C.13 Compute the determinants of: $\begin{bmatrix} 5 & \sqrt{3}+i \\ \sqrt{3}-i & 4 \end{bmatrix}$, $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 3 \\ 3 & 3 & 3 \end{bmatrix}$, and $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

Solution. Solution P.C.13

□

P.D.5 Use mathematical induction to prove that

$$1 + z + z^2 + \cdots + z^{n-1} = \frac{1 - z^n}{1 - z}$$

Proof. Proof P.D.5

□